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College of Science and Humanities-Jubail
Computer science Department



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Academic Year (2025-2026) - First Semester

ElImuna App



Version 1
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This Software Project Management Plans (SPMP) was prepared and provided as a deliverable for [Software Engineering, CS 411, 2025-2026] and it will be used by all Ellmuna Team, Students and graduates . This document is based in part on the IEEE Recommended Practice for SPMP Descriptions.

Team Members:

Table 0-1 Team Information

#	Name	ID	Role
1	Remas Alshehri	2220005349	Leader
2	Joud Alotaibi	2230006186	Member
3	Jori alghamdi	2230002748	Member
4	Jood Alali	2230004483	Member
5	Raya AL-harbi	2230005278	Member
6	Raghad Alasfoor	2230003688	Member

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Revision History

Table 0-2 Revision History

Name	Date	Reason For Changes	Version
All members	Oct. 12, 2025	Prepared initial version	0.1
All members	Oct. 13, 2025	Updated section 1	0.2
All members	Oct. 14, 2025	Updated section 2	0.3
All members	Oct. 15, 2025	Updated section 3	0.4
All members	Oct. 16, 2025	Updated section 4	0.5
All members	Oct. 17, 2025	Updated section 5	0.6
All members	Oct. 18, 2025	Complete review - Final version	1.0

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1. Project Overview

This section provides an overview of the **Ellmuna** App, describing its core purpose, scope, and objectives. It outlines the essential assumptions, constraints, and potential risks related to the project. Furthermore, it presents a detailed plan covering our deliverables, timeline, and budget, serving as a roadmap throughout the software development process. The section also includes a helpful list of key references, acronyms, and definitions to ensure clarity and consistency across all project documentation.

1.1 Purpose, Scope, and Objectives

The purpose of this project is to develop a unified application that offers diverse academic services for various educational levels in Saudi Arabia, ranging from High School to Postgraduate levels. The application supports students in calculating their academic results at a specific level. Also, It includes things like discovering universities and providing majors for master's and PhD levels. Therefore, the application is designed specifically for their academic needs.

The project solves many problems. For instance, when high school and bachelor's students want to know their percentage grade, there are many websites and applications. Each one of them has a different calculation, which might make the students stressed or confused. Postgraduates may face some issues when searching for universities that offer programs suitable for their needs. All that is considered a disadvantage; therefore, we developed our application to help them meet their goals.

The scope of our project is to develop an application that has four sections:

- 1- High school: In this part, the weighted grade will be calculated by entering high school GPA, Qudrat, and Tahsily. Then, it will be shown the academic grade, e.g., Excellent, Good, Pass, etc., as well as a message that will be written on it, the percentage of possibility of being accepted in Saudi universities.
- 2- Bachelor Students: the GPA will be calculated first by choosing whether it is out of 5 or 4. Then, enter the past credit hours and points, and current credit hours and points. After that, the GPA will be shown with a message if they are 1st honor, 2nd honor, or a message warns them.
- 3- Master student: it will be shown Saudi universities that offer master's programs, so they can select any university and only view the available majors.
- 4- PhD student: same way as the master's section, it will view the university and its majors only for PhD programs

The main objectives of the project are as follows:

- Unifying academic service into an application that serves all educational levels.
- Enable high school and bachelor's students to perform accurate GPA calculations easily and quickly.
- Help postgraduate students to have access to up-to-date information about universities and available majors for master's and PhD.
- Simple and interactive interface.
- Supporting the digital transformation of academic services in line with the Kingdom's Vision 2030 to develop education.

The project's objectives will be verified through performance tests, user experiences, and comparison of results with official formats adopted by Saudi universities. All functional and non-functional project requirements report (SRS) version 1.0 for the first semester of the 2025/2026.

1.2 Assumptions, Constraints and Risks

Table 0-3 Assumptions, Constraints and Risks

Type	Description
Assumptions	• All team members will collaborate and work as an effective team. • Each team member will have clearly defined roles and responsibilities to enhance the workflow. • All the calculations and the information about the universities and their majors should be accurate and should be taken from official resources. • Users should have a good connection to the internet while using the application. • The mathematical equation for calculating GPAs will not be changed during the project development.
Constraints	• The project must be completed during the first semester, which limits the possibility of adding advanced features outside the specified scope. • There is no actual budget allocated for the project. • The use of resources is limited to the personal computers available to the development team. • The quality of the application result depends on the accuracy of the data entered by users. • open-source libraries will be used to build the user interface (UI). • The application will be integrated with tools and math libraries when needed. • The technology to be used is Figma for UI/UX design • The application may be linked to the portals of Saudi universities via APIs after obtaining the necessary permits in the future.
Risks	• Team members may experience miscommunication, which will lead to misunderstanding among them. • Lack of technical experience among the development team in some programming stages may delay the schedule. • Error in the data entry committed by users, which leads to an inaccurate result. • It may be difficult to obtain updated data for all Saudi universities at the same time. • Admission policies might change at some universities, requiring periodic updates to the application.

1.3 Project Deliverables

Table 0-1 Project Deliverables

No.	Name of Deliverables	Submission Date	Media
1	Project Proposal	(21- 25 Sep)	Softcopy through Blackboard
2	Project Management Plan (SPMP)	(12 - 18 Oct)	
3	project status report 1	(19 - 25 Oct)	
4	Project Requirements (SRS)	(02 - 08 Nov)	
5	Project Design (SDS)	(09 - 15 Nov)	
6	Project Test Plan (STP)	(30 Nov - 04 Dec)	In-person meeting
7	Projects Presentation	(14 - 18 Dec)	

1.4 Schedule and Budget Summary

In this project, there is no budget because the development depends on personal resources (a *personal computer, a personal network, and free software tools*).

The table below is the schedule of major activities and their duration:

Table 0-2 Schedule Summary

Major Activity	Duration
Analysis Requirements	2 weeks
Design stages	3 weeks
Development stage	3 weeks
Test Evaluation stage	4 weeks
Documentation and Final Delivery stage	3 weeks

1.5 Evolution of the Plan

The software project management plan will follow the IEEE Std 1058-1998 standard.

The plan will be updated into two phases:

1. Scheduled updates: after completing each stage, i.e., (*analysis, design, development, test*), the plan is reviewed to document progress and adjustments.
2. Unscheduled updates: any unexpected changes that appear in the project scope, schedule, or requirements.

The initial version of this plan will be stored on the project's GitHub platform to ensure that modifications are stored and traceable. Moreover, publishing any official updates in the plan through downloading the latest version into the store ensures that it includes the verified issue number, date, and other feedback. Therefore, the supervisor is responsible for reviewing and approving any amendments to the plan content before it is adopted. So, after the approval, the new version is considered valid and officially adopted as a replacement for the previous version.

1.6 References

1. *IEEE Standard for Software Project Management Plans, Software Engineering Standards Committee of the IEEE Computer Society, IEEE-SA Standards Board.* (1998). <https://wildart.github.io/MISG5020/standards/IEEE-1058-1998.pdf>
2. Ruas, T., & Grosky, W. I. (2020, January 1). *Software Engineering: A Practitioner's Approach, 9th Edition*. Research Gate.
https://www.researchgate.net/publication/365946272_Software_Engineering_A_Practitioner
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<https://www.slideshare.net/slideshow/software-project-management-plan-233515740/233515740#1>

1.7 Definitions and Acronyms

Table 0-3 Terminologies

Terminology	Definition
IEEE standard	a common standard template that organizations and individuals with various technological backgrounds use for documentation.
Agile methodology	A flexible, iterative software development and project management approach that delivers small, incremental updates (called sprints) with ongoing feedback and adaptation to change.

Table 0-4 Definitions and Acronyms

Acronyms	Meaning
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
SDS	Software Design Specification
STP	Software Test Plan
GUI	Graphical User Interface
IDE	integrated development environment
UML	Unified Modeling Language
IEEE Standard	Institute of Electrical and Electronics Engineers
UI/UX	User Interface & User Experience
GPA	Grade Point Average
PhD	Doctor Of Philosophy
CGPA	Cumulative Grade Point Average
IAU	Imam Abdulrahman Bin Faisal University
WBS	Work Breakdown Structure
API	Application Programming Interface
STS	Software Test Sepecification

SDLC	Software Development Life Cycle
RTM	Requirements Traceability Matrix
CCB	Change Control Board
CRF	Change Request Form
SV	Schedule Variance
CV	Cost Variance
CPI	Cost Performance Index
EV	Earned Value
QA	Quality Assurance
AWS	Amazon Web Services
DBA	Database Administrator
SQL	Structured Query Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
PDF	Portable Document Format
MS	Microsoft
RAM	random access memory
GB	Gigabyte
QA	Log Quality Assurance Log
PM	Project Manager
GD	Google Drive
Docs	Google Docs
MS	Teams Microsoft Teams
DMAIC	Define – Measure – Analyze – Improve – Control
QA	Lead Quality Assurance Lead

1.8 Document Structure

This document consists of a set of sections that explain the various administrative and technical aspects of the project, and has been prepared according to the structure recommended in IEEE Std 1058-1998, as follows:

Section 1: Project Overview:

Provide a general introduction to the project, explaining its purpose, scope, and objectives, as well as the assumptions, constraints, risks, deliverables, references, and definitions.

Section 2: Project organization:

It clarifies the organizational structure of the project and the relationship between stakeholders, such as developers and the academic supervisor. Moreover, it defines the role and responsibilities of each member in the project.

Section 3: Managerial Process Plans:

Explain the project management plans in terms of the plan for starting the project, schedule, resource allocation, follow-up on progress, managing risks, and the project close-out plan.

Section 4: technical process plans:

Explains the technical processes for application development, which is Agile, the tools and methodologies used, the technical infrastructure, and product acceptance criteria.

Section 5: supporting process plans:

It outlines project support processes such as documentation, quality assurance, performance monitoring, code review, and backup.

Section 6: Additional plans:

All additional plans that are related to information security, privacy, user support, and plans for maintenance or expansion of the project in the future.

2. Project Organization

2.1 External Interfaces

The "علمنا – Ellmuna" project is one of the student projects in the Computer Science Department at Imam Abdulrahman bin Faisal University (IAU). It is a Software Engineering Project course project and is supervised by Dr. Noorah Al-Behaijan. The university serves as the main umbrella organization, where the department represents the academic university, we operate through. Dr. Noorah acts as the supervising instructor/customer who naturally evaluates our overall progress and the final outcomes.

Our team operates entirely within the department; we haven't involved any external companies or collaborators. However, we did review some familiar sites and apps, e.g., Moadley and Qabol but for a conceptual only. We did this just to better our design concepts and make sure that we offer so much more user experience.

Who Are the Users? the target audience of "علمنا – Ellmuna" are students at various school levels in Saudi Arabia. Below is a categorization:

High School Students, the application helps them by providing two services:

1. They use the application to accurately calculate their weighted admission score from their high school percentage, aptitude test, and achievement test. Once they enter their grades, the system immediately shows the final weighted score, along with a simple descriptive message that helps reflect the student's standing based on that result. The app also helps them estimate their potential acceptance into Saudi universities; this is crucial for planning ahead and making much better academic decisions.
2. Saudi University Exploration (bachelor's degree): They can surf and explore Saudi universities, view colleges and courses available, and Students can compare programs at different universities without wasting time researching each university and its offerings. This reduces distraction and the proliferation of resources and helps organize all their requirements through a single application that brings all universities together for easy access.

Undergraduate students: the program accurately calculates a student's semester and cumulative GPA by requesting some basic information from the student, such as their university's GPA of 4.0 or 5.0, based on the scale chosen by the student. With the student's previous hours completed, the application can easily calculate their semester and cumulative GPA. After calculating their GPA, a message is displayed explaining the student's academic status. The student receives a message indicating whether they are on an academic warning, honor roll, or excellent performance, based on their cumulative GPA, making it easier to track their academic progress.

Graduate students (master's and doctoral): It allows them to view the universities in Saudi Arabia that offer graduate studies, and they can view the programs offered by each university, without having to search each university's website to find out the studies available there, their type, and the programs they offer. This feature supports fast and convenient searching and comparison between universities and academic programs within their chosen fields.

"**علمنا – Ellmuna**" successfully meets the needs of students from high school right up to postgraduate studies by combining academic calculation tools with informational browsing features in one smart, unified, and very user-friendly platform.

Communication and Coordination

University email and classroom meetings facilitate communication with our supervisor, Dr. Noura Al-Bahjan, when needed through our team leader, Rimas Al-Shehri. Regarding communication between our team members and our team leader, we are in constant touch. We hold a weekly Zoom meeting every Tuesday for one or two hours, depending on the need. Everyone is eager to attend to discuss project progress, allocate tasks, and foster a spirit of collaboration among us as a team. We also have a dedicated WhatsApp group for quick daily communication and sharing the latest work updates and announcements.

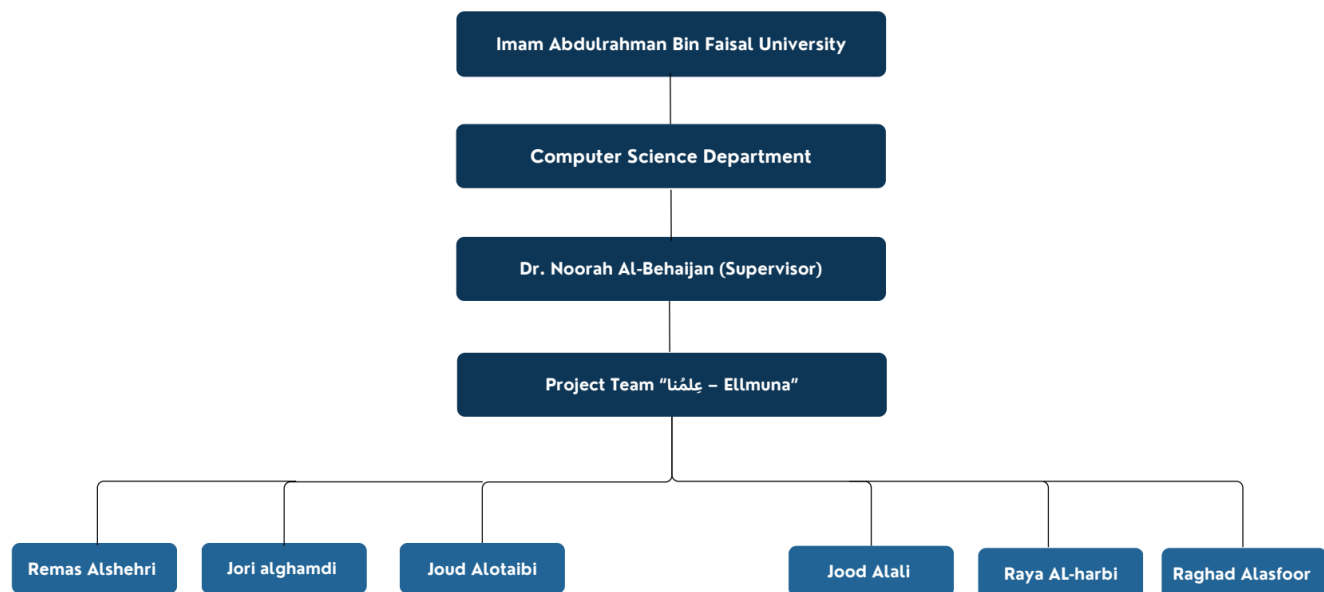


Figure 1 External Organizational Chart showing the relationship

2.2 Internal Structure

Our "علمنا – Ellmuna" team consists of six members from the Computer Science Department. The team leader is Reemas Al-Shehri, and as a team leader, she is responsible for communicating with the supervisor if necessary, additionally she organizes the work for the project and assigning tasks to each member, arranging meetings between us, and ensuring the project is delivered within the delivery period. Every team member has her own task and responsibility, like analysis, design, implementation, testing, documentation, and other responsibilities required to complete the project. Although each of us is individually accountable for our work, we also verify one another's work as the final step before submitting any final work to ensure the quality and integrity of our work. We are in constant contact through a WhatsApp group and we have weekly Zoom meetings every Tuesday to review progress and distribute new assignments by the leader and we use Google Drive to share documents and maintain synchronized records. Each member must complete her part first; then we all as a team review work one by one to ensure consistency, quality, and completeness before submission to the supervisor.

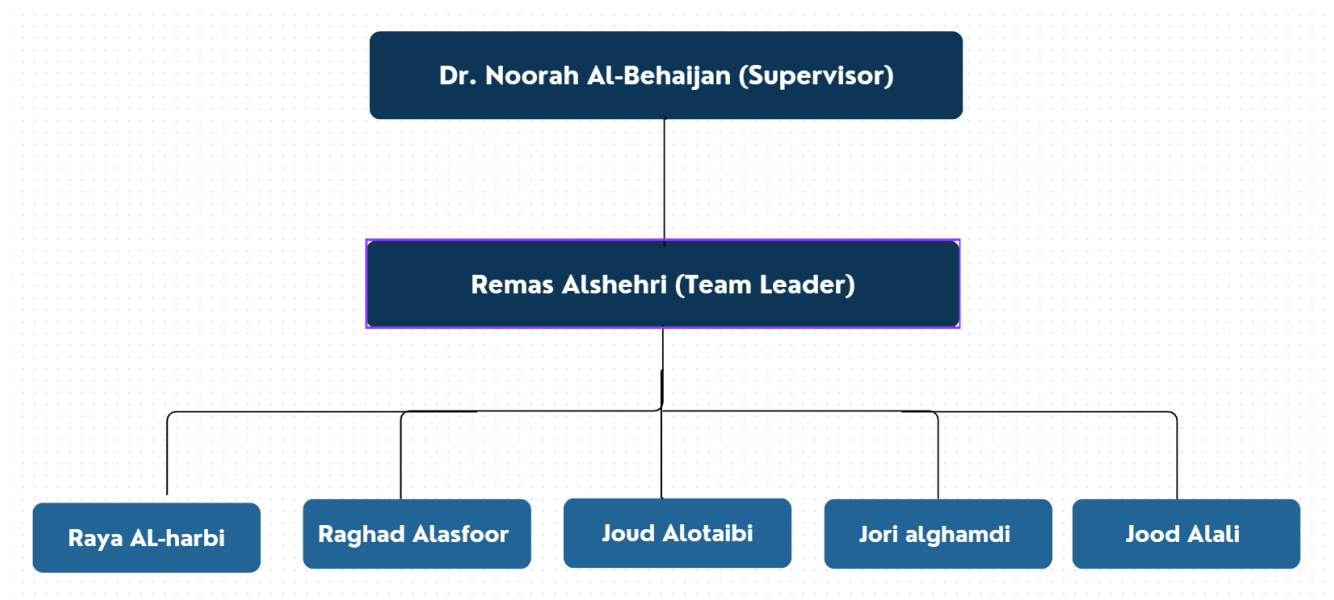


Figure 2 Internal Organizational Chart and Communication Hierarchy

2.3 Roles and Responsibilities

Our "علمنا – Ellmuna" team organized the work to make sure every single member contributes across the different project stages from the initial concept right up to the final documentation. This approach ensures the effort is balanced and keeps the team well-integrated.

Team Roles and Main Responsibilities

Table 0-6 Team Roles and Main Responsibilities for the "علمنا – Ellmuna" Project.

No.	Name	Main Responsibilities
1	Remas Alshehri	Project Coordination and Supervision: This member coordinates project activities, allocates tasks, plans meetings, reads documents, and coordinates with the supervisor.
2	Raya Al-Harbi	Documentation & Support: This member assists in documentation and report preparation, works on design and test tasks, and assists in preparing the final presentation.
3	Raghad Al-Asfoor	System Analysis & Core Implementation: This role is critical. It involves running the system analysis, applying the GPA/CGPA calculation logic, and then testing the accuracy of those results very carefully.
4	Joud Al-Otaibi	UI/UX Design & Visuals: This person is responsible for the actual UI/UX design of the user interface and overall user experience, including helping with the application's general visual coordination.
5	Jori Al-Ghamdi	Technical Document Preparation: This role is all about the paperwork. It includes preparing and constantly updating all technical documents (Idea, Proposal, SPMP, Final Report) and taking an active part in every team review.
6	Jood Al-Ali	Testing & Presentation: This member takes the lead on conducting all system testing, documenting the results, and assists the entire team with preparing the final presentation.

Work Activity Matrix

Table 0-7 Work Activity Matrix for the “علمنا – Ellmuna” Project.

Project Activity	Remas Alshehri	Raya Al-Harbi	Raghad Al-Asfoor	Joud Al-Otaibi	Jori Al-Ghamdi	Jood Al-Ali
Requirements Analysis	✓	✓	✓			
System and UI/UX Design	✓			✓		
Implementation and Coding			✓			
System Testing	✓	✓				✓
Documentation	✓	✓			✓	
Final Review	✓	✓	✓	✓	✓	✓
Presentation Preparation	✓	✓				✓

We always make it a priority to go through all our deliverables as a group before any submission. This ensures both accuracy and quality. Every single member is actively engaged in writing, reviewing, and refining that final report it's a collective effort.

3. Managerial Process Plans

This section of the Project Management Plan specifies the project management processes for the project. This section defines the plans for project start-up, risk management, project work, project tracking, and project close-out.

3.1 Start-up Plan

3.1.1 Estimates

The estimation process for the **Ellmuna** project focuses on predicting the time, resources, and effort required to complete all project phases — from analysis to presentation. Since this is a student academic project, no financial costs are involved other than the use of personal devices and internet access.

Estimation Methods and Sources

We employ a combination of estimation techniques to ensure accuracy:

- Analogy: Comparing with similar past projects
- Rule of Thumb: Quick, experience-based estimates
- Expert Judgment: Leveraging team members' expertise

These methods are supplemented by research and expert-assisted reviews to provide a comprehensive understanding of the project's requirements.

Cost Estimates

Since **Ellmuna** is a student academic project, it does not involve direct financial expenses. All work is completed using personal devices and freely available tools.

Development Cost: No direct cost; students use personal laptops, mobile devices, and home internet connections.

Software Cost: All tools used (Android Studio, Figma, Firebase, GitHub, Google Drive) are free or have student licenses.

Operational Cost: Negligible, as the work is conducted remotely and through university resources.

Basis for Estimation: The cost estimation follows a zero-budget model since no external funding or paid services are required.

Confidence Level: 100%, as all tools are confirmed free and accessible.

Schedule Estimates

The project is expected to run from September 25, 2025, to December 19, 2025, covering 12 weeks of academic work. The schedule is divided into short iterations (sprints), each lasting approximately two weeks, with specific deliverables in each.

Table 0-9 Schedule Estimates

Phase / Iteration	Estimated Duration	Main Deliverable
Requirement Analysis	2 weeks	SRS Document
System & UI Design	2 weeks	Design diagrams, UI mockups
Implementation	4 weeks	Functional Prototype
Testing	2 weeks	Test Report
Documentation & Presentation	2 weeks	Final Report & Presentation Slides

Confidence Level: 80%, as small delays may occur due to academic workload.

Resource Requirements

The **Ellmuna** project requires human, hardware, and software resources as listed below:

- Human Resources: 6 team members (Team Leader, Developers, Designer, Document Specialist, Tester).
- Hardware: Personal laptops (≥ 8 GB RAM) and Android phones for testing.
- Software: Android Studio, Firebase, Figma, GitHub, Google Drive, MS Word.
- Network: Stable internet for cloud synchronization and online collaboration.

Confidence Level: 90%, based on available student resources and prior experience.

Re-estimation Process

Re-estimation will occur at key milestones or if major changes happen in project requirements.

The team will:

1. Review progress at the end of each sprint (every 2 weeks).
2. Compare actual time spent versus planned time.
3. Update the resource and schedule plan if necessary.
4. Discuss changes with the supervisor before final approval.

Re-estimation will use the same methods as the original estimation analogy from previous projects and WBS breakdown.

Technical Specifications

The project is developed as a mobile application using:

- **Programming Languages:** Java and Dart (Flutter Framework).
- **Database:** Firebase Realtime Database and Firestore queries.
- **Development Environment:** Android Studio and Visual Studio Code
- **Design Tools:** Figma and Draw.io for UI and architecture diagrams.
- **Version Control:** GitHub for managing project updates.

This technical setup ensures efficient collaboration, fast iteration, and reliable testing throughout the development process

3.1.2 Staffing

The **Ellmuna** project is entirely designed and implemented by six internal personnel in the Computer Science Department, College of Science and Humanities – Jubail, Imam Abdulrahman Bin Faisal University. There will be no external hiring or subcontracting. All resources are internal; students engaged with the team of contributors.

Each member of the team is involved in every phase of the project. Roles are based on the individual skill level with a promise of flexibility in cross-support. Each person engages in both technical and management activities to ensure shared responsibility and equitable workloads. The table below indicates the number of personnel, skill level, and time allocated to each role.

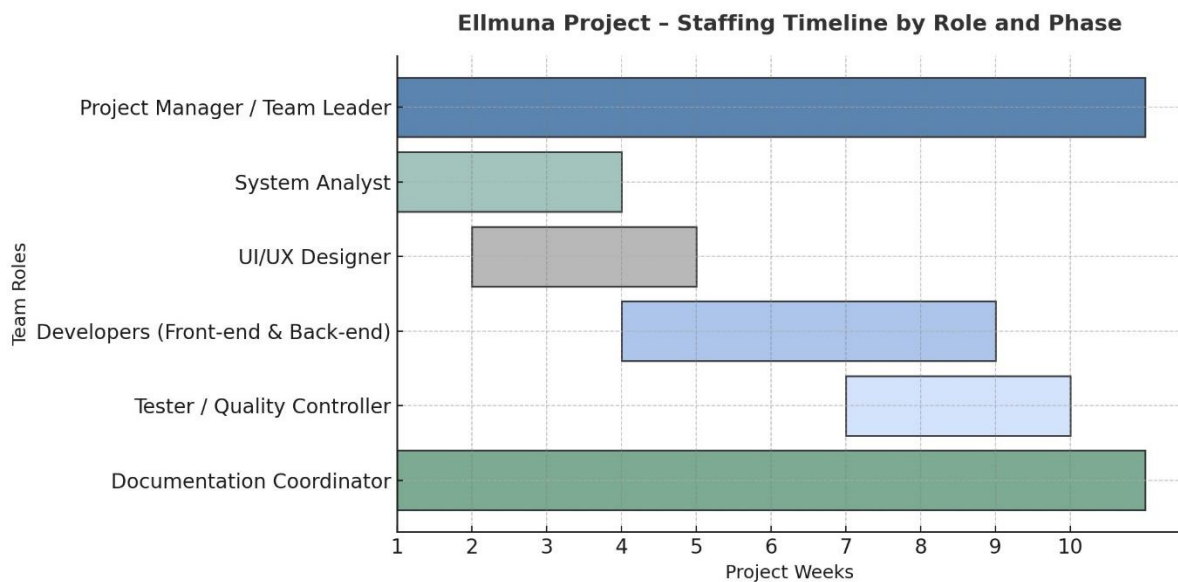


Figure 3 Ellmuna Project Staffing

Required Staff

Our project requires a total of 6 personnel with varying skill levels:

Table 0-10 Role and Skill level

Role / Skill Level	Description of Responsibilities	No. of Members	Skill Level	Project Phase(s)	Duration (Weeks)
Project Manager / Team Leader	Plans, coordinates, and supervises all project activities, communication, and scheduling.	1	High	All phases (Planning to Closeout)	1–10
System Analyst	Defines functional and non-functional requirements; documents use cases.	1	Medium	Initiation & Planning	1–3
UI/UX Designer	Creates user-interface prototypes and ensures usability and accessibility.	1	Medium	Design & Implementation	2–4
Developers (Front-end & Back end)	Implement features, connect UI with database, and maintain codebase.	2	High	Implementation & Testing	4–8
Tester / Quality Controller	Performs verification, validation, and usability testing; tracks issues.	1	Medium	Testing & Quality Assurance	7–9
Documentation Coordinator	Prepares and reviews all project documents (SPMP, reports, presentations).	1	Medium	All phases (especially Closeout)	1–10

Resource Allocation Visualization

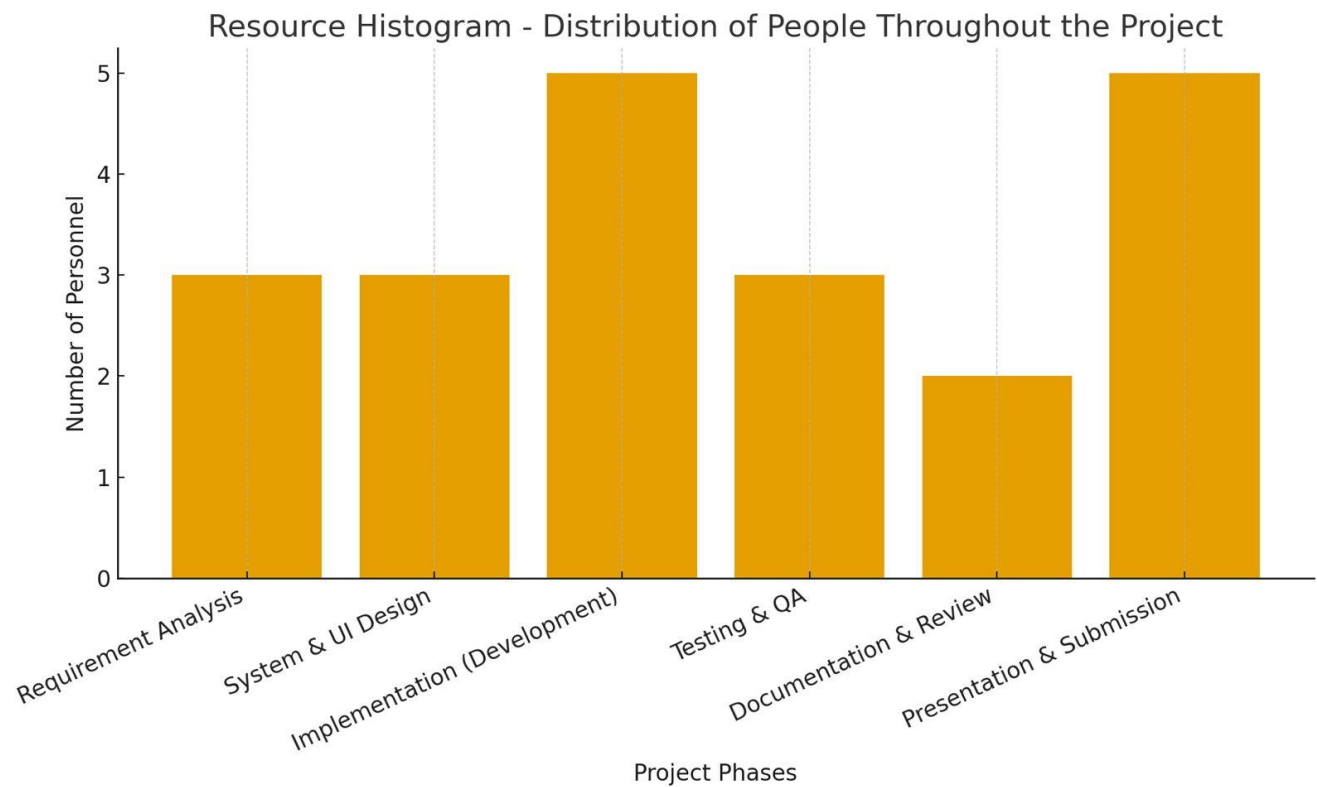


Figure 4 Histogram show people throughout

3.1.3 Project Staff Training

A structured training plan will be implemented at the beginning of the **Ellmuna** project and reinforced at each major phase. The training plan will ensure that all team members have the technical, managerial, and supporting skills that will relate to completing the project objectives as efficiently as possible.

The training will ensure that all roles in the team (project manager, analyst, designers, developers, tester, and documentation coordinator) will have enough knowledge of and capabilities to perform within their role effectively.

Table 0-11 Project Staff Training Overview

Training Type	Purpose	No. of Personnel	Method	Skill Category
Technical Tools Training	Enhance technical competence in Android Studio, MySQL, Figma, and GitHub for collaborative development.	6	Computer-assisted training and peer mentoring	Technical
Project Management & Communication	Improve coordination, planning, and scheduling skills using Trello, Gantt charts, and communication tools.	6	Interactive lectures and consultations	Managerial
Testing & Quality Assurance	Train on writing test cases, performing functional and usability testing, and using bug-tracking sheets.	6	Mentoring and hands-on workshops	Technical
Documentation & Reporting	Develop professional reporting, formatting, and adherence to IEEE standards for SPMP and other documents.	6	Guided practice and collaborative peer review	Supporting
Team Collaboration & Leadership	Strengthen teamwork, leadership, and problem-solving skills for effective group decision-making.	6	Group discussions and scenario-based learning	Managerial / Supporting

Training will take place entirely within the project team. The project manager will schedule weekly short meetings to share knowledge among team members, and experienced ones will mentor other members when a specialized skill is needed. The training will mostly take place in the initiation and planning phases of the project (Weeks 1-3), with refresher training in phases 2 and 3 to ensure technical and managerial practices are consistent. Upon completion of the training, all six members will have the appropriate technical skills, including communication and documentation skills, to successfully complete the **Ellmuna** project.

3.2 Work Plan

3.2.1 Work Breakdown Structure

The Work Breakdown Structure (WBS) decomposes the **Ellmuna** project into manageable work activities. Each activity identifies the necessary resources, duration, deliverables, acceptance criteria, and dependencies (predecessor and successor). This breakdown helps to expose risk factors and ensure accurate estimation for resources and scheduling.

Table 0-12 Work Breakdown Structure

Levels	Phase	Description	Resources Needed	Estimated Duration	Deliverables	Acceptance Criteria	Predecessor / Successor
0.1	Requirement Analysis	Collect and analyze user needs, define main features (GPA & Weighted Score Calculators, University Guide).	Team laptops, Google Docs, Internet, Course Guidelines	2 weeks	SRS Document	Requirements approved by supervisor	None / Design Phase
0.2	System & UI Design	Create architecture diagrams, database schema, and UI mockups.	Draw.io, Figma, UML tools	2 weeks	SDS Document, UI Mockups	Design aligns with requirements and passes peer review	Analysis / Implementation
0.3	Implementation (Development)	Develop front-end and back-end modules for calculators and database.	Visual Studio Code, Database tools	4 weeks	Working Prototype	Core functions run without major bugs	Design / Testing
0.4	Testing Phase	Conduct unit, integration, and user testing.	Testing tools, sample data	2 weeks	STS Report, Bug Fix Log	Almost all test cases passed	Implementation / Documentation
0.5	Documentation & Review	Prepare SPMP, Final Report, and User Manual.	MS Word, Google Drive	1 week	Final Documents	Supervisor approval	All previous phases
0.6	Presentation & Submission	Final presentation, demo, and submission to Blackboard.	PowerPoint, Recorded Demo	1 week	Presentation Slides, Demo Video	Supervisor approval and positive feedback	Documentation / Project Close

Risk Factors:

- Incomplete requirements may delay the design phase.
- UI design not meeting user expectations could require rework.
- Bugs found during the testing phase may impact deployment timing.

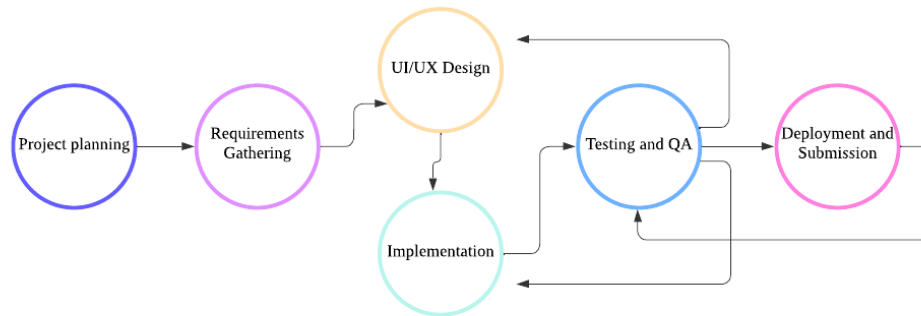


Figure 5 PERT CHART FOR WBS

3.2.2 Schedule Allocation

The schedule for the **Ellmuna** project was designed to ensure an organized workflow and timely completion of all phases. The activities are scheduled in sequence with partial overlaps to optimize time while maintaining quality. This section outlines the main scheduling relationships, dependencies, and project milestones. The table below summarizes the time allocation and sequencing of the main project phases:

Table 0-13 Schedule Allocation

Phase	Start Date	End Date	Duration
Requirement Analysis	Sep 25	Oct 08	2 weeks
System & UI Design	Oct 09	Oct 22	2 weeks
Implementation	Oct 23	Nov 19	4 weeks
Testing	Nov 20	Dec 03	2 weeks
Documentation & Review	Dec 04	Dec 10	1 week
Presentation & Submission	Dec 11	Dec 17	1 week

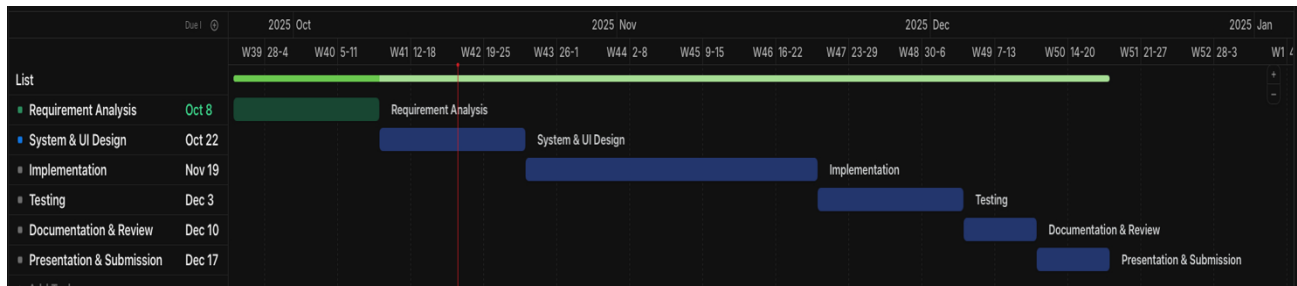


Figure 6 Ellmuna Project Schedule

Critical Path

The critical path for the **Ellmuna** project is as follows:

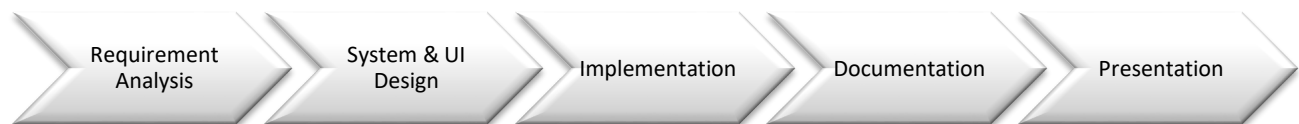


Figure 7 The critical path

Milestones

1. Project Proposal Approved – Week 1
2. Design Completed – Week 4
3. Prototype Functional – Week 8
4. Testing Completed – Week 10
5. Final Report and Presentation Delivered – Week 12

Constraints

- Supervisor approval is required before proceeding from Analysis to Design.
- The final presentation date is fixed by supervisors.
- Documentation and Testing phases overlap slightly to save time.

3.2.3 Resource Allocation

This section defines the distribution of all humans, technical, and administrative resources required for each major work activity in the project. It specifies the skill levels, computing resources, software tools,

testing and simulation facilities, and administrative support assigned to each activity according to the Work Breakdown Structure (WBS).

The following table outlines the resources allocated to each major work activity in the Work Breakdown Structure (WBS) for the project. Resources include personnel by skill level, computing resources, software tools, and administrative support.

1. Requirement Analysis Phase

Objective: Identify and document user needs, system goals, and project scope. Duration: 2 weeks

Table 0-1 Requirement Analysis Based on Resource type (Requirement Analysis Phase)

Resource Type	Specific Resource	Quantity	Skill Level / Role	Allocation Purpose
Personnel	Project Manager	1	Advanced (project planning, risk control)	Coordinate communication and approve requirement documents (SRS).
Personnel	System Analyst	1	Senior (requirement engineering, business analysis)	Collect requirements, analyze user needs, define functional scope.
Personnel	Technical Writer	1	Intermediate (documentation skills)	Draft requirement specifications and meeting summaries.
Computing Resources	Laptops / Desktops	3	Standard specs (8 GB RAM, Core i5)	Used for documentation and online collaboration.
Software Tools	Google Docs, Google, Meet, Microsoft Teams	-	Basic office tools	Requirement collection, document, review, and team meetings.
Administrative Support	File Organization Assistant	-	Clerical	Organize project files and meeting minutes.

2. System & UI Design Phase Objective: Design architecture diagrams, database schema, and interactive user interface prototypes. Duration: 2 weeks

Table 0-2 Requirement Analysis Based on Resource type (System & UI Design Objective) Phase

Resource Type	Specific Resource	Quantity	Skill Level / Role	Allocation Purpose
Personnel	System Architect	1	Senior (software architecture, database design)	Create logical and physical architecture diagrams.
Personnel	UI/UX Designer	1	Intermediate (UX wireframing, graphic design)	Design interactive user interface mockups.
Personnel	Project Manager	1	Advanced	Validate design decisions and ensure requirement alignment.
Computing Resources	High-spec Design Workstations	2	16 GB RAM, Core i7, dedicated GPU	Used for UI and diagram tools (e.g., Figma, Draw.io).
Software Tools	Figma, Lucidchart, Draw.io, Visual Paradigm	-	Design and modeling tools	Create UI mockups and UML models..
Administrative Support	SDS Documentation Assistant	-	Clerical	Assist in formatting System Design Specification document.

3. Implementation (Development) Phase Objective: Develop, integrate, and test application modules. Duration: 4 weeks

Table 0-3 Requirement Analysis Based on Resource type - Implementation (Development) Phase

Resource Type	Specific Resource	Quantity	Skill Level / Role	Allocation Purpose
Personnel	Front-end Developer	1	Intermediate (JavaScript, React Native / HTML CSS)	Implement user interface and navigation components.
Personnel	Back-end Developers	2	Senior (Python Django, API integration)	Develop business logic, GPA calculations, and database services.
Personnel	Database Administrator (DBA)	1	Intermediate (PostgreSQL, SQL optimization)	Create and maintain relational database schema.
Personnel	Project Manager	1	Advanced	Monitor progress and track tasks in Jira.
Computing Resources	Developer Workstations	4	Core i7, 16 GB RAM	Used for code development and testing.
Software Tools	VS Code, GitHub Pro, Jira, Docker	-	Professional IDE and CI/CD tools	. Version control and agile tracking.
Software Tools	Django Framework, PostgreSQL	-	Open-source backend tools	Core development and database environment.
Special Testing Facilities	AWS EC2 Instance, Local Staging Server	-	Cloud deployment platform	Host test versions for integration testing.
Administrative Support	Documentation Coordinator	-	Clerical	Track development logs and change requests.

4. Testing & Quality Assurance Phase Objective: Validate system performance, reliability, and user satisfaction. Duration: 2 weeks

Table 0-4 Requirement Analysis Based on Resource type - Testing & Quality Assurance Phase

Resource Type	Specific Resource	Quantity / Staff	Skill Level / Role	Allocation Purpose
Personnel	QA Engineer	1	Intermediate (test planning, bug tracking)	Prepare and execute test cases, report defects.
Personnel	Developers (for bug fixes)	2	Intermediate	Fix issues identified in testing reports.
Computing Resources	QA Workstations	2	8 GB RAM, Core i5	Used for test execution and recording.
Software Tools	Selenium, TestRail, Postman	-	Automated and manual testing tools	. Used for test automation and API validation.
Special Testing Facilities	AWS Testing Environment		Cloud-based simulation	Used for load and performance testing.
Administrative Support	Test Report Coordinator	-	Clerical	Organize test logs and final reports.

5. Documentation & Review Phase Objective: Prepare, review, and finalize all project deliverables and reports (SPMP, SRS, SDS, STS, User Manual). Duration: 1 week

Table 0-5 Requirement Analysis Based on Resource type - Documentation & Review Phase

Resource Type	Specific Resource	Quantity / Staff	Skill Level / Role	Allocation Purpose
Personnel	Technical Writer	1	Intermediate (document editing)	Draft and format technical reports.
Personnel	Project Manager	1	Advanced	Approve final deliverables and validate content
Computing Resources	Laptops / Office PCs	2	Standard spec	Used for document editing and storage.
Software Tools	MS Word, Google Docs, Grammarly	-	Office suite and review tools	. Create and edit project documents.
Administrative Support	Print and File Coordinator	-	Clerical	Prepare digital and hard-copy versions.

6. Presentation & Submission Phase Objective: Prepare and present the final project to stakeholders and academic evaluators. Duration: 1 week

Table 0-6 Requirement Analysis Based on Resource type - Presentation & Submission Phase

Resource Type	Specific Resource	Quantity / Staff	Skill Level / Role	Allocation Purpose
Personnel	All Team Members	4	Intermediate	Prepare slides and demo presentation.
Personnel	Project Manager	1	Advanced	Lead presentation and Q&A session.
Computing Resources	Presentation Laptop	1	High spec	Used for demo and slide show.
Software Tools	Microsoft PowerPoint, Canva Pro	-	Design and presentation tools	Develop slides and visuals.
Special Facilities	Camera, Microphone, Projector	-	AV equipment	Record and deliver presentation.
Administrative Support	Submission Assistant	-	Clerical	Handle file uploads and final submission.

The resource allocation ensures the project has adequate human, computational, and administrative resources throughout all life-cycle phases. Personnel are distributed according to specialization, while computing and software tools are reused efficiently to reduce cost and overlap.

3.2.4 Budget Allocation

This section provides a detailed financial breakdown of the resources required for each major activity identified in the (WBS). The budget covers human resources, hardware, software tools, computing resources, meetings, travel, testing, cloud services, and administrative support. Each itemized entry represents a specific cost type associated with the respective project phase. All estimates are based on local Saudi market rates and typical academic software project environments.

This table details the budget for each WBS activity, estimated in free. Costs cover personnel meetings, computing resources, software tools, and testing facilities.

Table 0-7 Budget Allocation

Work Activity	Resource Type	Description	Cost (SAR)
Requirement Analysis	Personnel	Team Lead (5 days), Requirements Engineer (5 days)	free
	Meetings	Virtual stakeholder meetings	free
	Computing Resources	Laptop and internet rental	free
	Software Tools	Google Docs, Visio subscriptions	free
	Administrative Support	Assistant (2 days)	free
Total			0.0 Saudi Riyal
System & UI Design	Personnel	UI Designer (5 days), System Architect (5 days)	free
	Computing Resources	Desktop rental	free
	Software Tools	Figma, Lucidchart subscriptions	free
Total			0.0 Saudi Riyal
Implementation (Development)	Personnel	Developers (10 days each), Module Lead (10 days)	free
	Computing Resources	Workstations, cloud server	free
	Software Tools	IDE, GitHub, Firebase	free
	Administrative Support	Assistant (3 days)	free
Total			0.0 Saudi Riyal
Testing Phase	Personnel	Test Engineers (5 days each)	free
	Computing Resources	Laptops, emulators	free
	Software Tools	Selenium, Jira	free
	Special Testing Facilities	Virtual environment setup	free

Total			0.0 Saudi Riyal
Project Documentation & Review	Personnel	Documentation Lead (3 days), Review Team (3 days)	free
	Computing Resources	Laptop usage	free
	Software Tools	Word, Grammarly	free
	Administrative Support	Assistant (1 day)	free
Total			0.0 Saudi Riyal
	Personnel	Project Manager (2 days), Team Lead (2 days)	free
	Computing Resources	File transfer tools	free
	Software Tools	Google Drive	free
	Administrative Support	Assistant (1 day)	free
Total			0.0 Saudi Riyal

3.3 Project Tracking Plan

3.3.1 Requirements Management

3.3.1.1 Overview

The Requirements Management Plan defines the methods and procedures for identifying, documenting, tracking, analyzing, and controlling all project requirements throughout the software development lifecycle (SDLC).

This plan ensures that all functional and non-functional requirements remain consistent, traceable, and aligned with the project's goals, schedule, budget, and quality expectations.

3.3.1.2 Requirements Tracking and Traceability

All requirements will be uniquely identified and documented within the Software Requirements Specification (SRS) document.

A Requirements Traceability Matrix (RTM) will be maintained to link each requirement to its corresponding design component, implementation module, and test case.

This enables:

- **Verification** that all requirements are implemented and tested.
- **Impact Analysis** when changes are proposed.
- **Accountability** by showing which requirement is responsible for each deliverable.
- deliverable.

Table 0-8 Requirements Tracking and Traceability

Requirement ID	Description	Design Element	Code Module	Test Case ID	Status
REQ-01	GPA Calculation Function	Function Module 1	calculate_gpa	TC-01	Implemented
REQ-02	Weighted Score Calculator	Function Module 2	weighted_score	TC-03	In Progress
REQ-03	University Recommendation Guide	Database + API	uniguide_api.py	TC-05	Planned

3.3.1.3 Change Control Process

The Requirements Change Control Process ensures that any modification to the project requirements is formally reviewed, analyzed, and approved before implementation.

Change Request Steps:

1. **Submission:** A team member or stakeholder submits a **Change Request Form (CRF)** describing the proposed modification.
2. **Recording:** The change is logged in the **Change Request Register** with a unique ID and date.
3. **Impact Analysis:** The project team evaluates the effect on:
 - a. Product scope and functionality
 - b. Schedule and milestones
 - c. Budget and resources
 - d. Risk factors and quality metrics
4. **Review & Approval:** The Change Control Board (CCB) reviews the analysis results and decides to:
 - a. Approve
 - b. Defer
 - c. Reject
 - d. Request further information
5. **Implementation & Tracking:** Approved changes are integrated into the SRS and related design documents and tracked until closure.
6. **Reporting:** The Project Manager updates the change log and communicates the change status during the weekly progress meeting.

3.3.1.4 Change Control Board (CCB)

A Change Control Board (CCB) will be established to oversee all proposed changes and ensure alignment with project objectives.

Table 0-9 Change Control Board (CCB)

Role	Member	Responsibilities
Project Manager (Chair)	Oversees change process, ensures documentation accuracy, and approves/rejects changes.	Lead CCB meetings, review change requests, ensure alignment with project objectives, and approve or reject proposed changes.
System Analyst	Evaluates the impact of the change on scope, design, and architecture.	Assess how proposed changes affect system requirements, data models, and integration points.
Developer Representative	Assesses technical feasibility and implementation effort.	Estimate the development effort, identify potential risks or conflicts in code, and suggest alternative technical solutions.
Quality Assurance Lead	Evaluates impact on testing, quality, and validation processes.	Ensure that changes do not compromise quality standards, update test plans, and validate system stability after changes.
Supervisor / Client Representative	Provides final decision on critical requirement changes.	Approve significant modifications, confirm that changes meet business goals, and ensure client satisfaction

CCB meetings will be held **bi-weekly** or upon submission of any major change request.

3.3.1.5 Requirements Modeling, Prototyping, and Reviews

To improve clarity and reduce ambiguity, requirements will be analyzed and validated using:

- a. **Prototyping:** Early visual or functional prototypes (UI mockups, flow diagrams) built using tools like *Figma* or *Draw.io* to demonstrate the expected behavior of system features.
- b. **Modeling:** UML diagrams (Use Case, Class, Activity) will represent system interactions and dependencies.
- c. **Review Sessions:** Conducted after each major requirement milestone with stakeholders to ensure agreement before proceeding to design.

This process helps stakeholders visualize requirements, reducing the likelihood of late-stage rework or misunderstanding.

3.3.1.6 Impact Analysis

Every proposed requirement changes triggers a formal Impact Analysis.

The analysis will determine how the change effects:

- **System architecture** (new components, data flow modifications)
- **Testing requirements** (new or updated test cases)
- **Development effort** (additional coding hours, resource shifts)
- **Schedule and Budget** (potential timeline or cost adjustments)
- **Risk Exposure** (introduction of new risks or escalation of existing ones)

The results of each analysis are recorded in the Change Impact Report and shared with all CCB members for review.

3.3.1.7 Configuration Management Integration

The requirements management process is fully integrated with the Configuration Management Plan, ensuring:

- All requirement changes are version-controlled using Git repositories.
- Each approved change is tagged with a version number and documented in the Change Log.
- Baselines (e.g., SRS v1.0, v2.0) are created after each major review or release.
- Historical versions of requirements are preserved for audit and rollback purposes.

3.3.1.8 Requirements Verification and Validation

Verification ensures that requirements are correctly implemented, while validation ensures that they meet stakeholder needs.

Table 0-10 Requirements Verification and Validation

Activity	Purpose	Responsible	Timing
Requirements Review	Validate completeness and clarity of SRS	System Analyst & Supervisor	After Analysis Phase
Prototype Review	Validate user interface and interaction flow	Developer & Client	During Design Phase
Traceability Verification	Ensure each requirement has matching design, code, and test linkage	QA Team	After Implementation
Final Validation	Confirm that all approved requirements are delivered	Supervisor	Before Submission

3.3.1.9 Reporting and Communication

Regular updates on requirements progress and changes will be shared through:

- **Weekly progress reports** summarizing approved and pending change requests.
- **Shared Google Docs & Sheets** for collaborative tracking of the RTM and CRF logs.
- **Review meetings** at each phase milestone to ensure consistent understanding across all team members and stakeholders.

3.3.1.10 Summary

The Requirements Management Plan ensures that:

- All requirements are clearly documented, traceable, and version controlled.
- Any change is properly evaluated for its impact before approval.
- Communication between developers, analysts, and clients is transparent.
- The project remains aligned with its original objectives in terms of scope, cost, quality, and schedule.

This structured and traceable process minimizes uncontrolled changes (“scope creep”) and enhances the overall predictability and quality of the project outcomes

3.3.2 Schedule Control

The **Ellmuna** project utilizes a structured schedule control method to keep tabs on progress, ensure timely key milestone completion, and ensure adherence to the approved project schedule. This method identifies the procedures, tools, and measurement criteria that will be used throughout the project's life cycle to measure, compare, and control schedule performance.

Progress Measurement: The Project Manager will monitor the status of work activities weekly based on measuring a set of major and minor milestones defined in the Work Breakdown Structure (WBS). Each task's percent completion will be recorded in a collective progress sheet, which will be reviewed during a team meeting to confirm that the task is keeping up with the planned schedule. Project tracking tools will be used to record updates and for visuals to show ongoing progress and delays, or dependencies, between tasks.

Planned vs. Actual Progress Assessment: At the end of each phase, actual progress will be measured against the baseline project schedule with Gantt charts and overall task completion reports. Once all activity records indicating the planned vs. actual schedule progress are updated, any variance to planned progress will be assessed to determine the cause of the delay—resourcing, technical difficulty, or dependency bottlenecks. The project manager will review the discussions with the team and amend the timeline or reallocate resources to recover the project schedule.

Corrective Actions: The team will take corrective action immediately whenever actual progress deviates from the planned progress. Examples include reassigning tasks, modifying due dates/timelines, adding support, or reprioritizing to get back on track. Any corrective actions will be documented in the Schedule Change Log to preserve a record of actions taken and for future reference.

Methods and Tools: The team will use Microsoft Excel and Google Sheets for progress tracking and schedule updates and may develop Gantt charts to visually show task sequences and dependencies. The central location for the up-to-date schedule documents will be Google Drive, allowing all team members access to needed shared materials. Weekly meetings via Microsoft Teams provide a forum for discussing group progress and raising any related concerns.

Definitive criteria for completion of each milestone will be based on clear and objective criteria to assess the scope and quality of work undertaken. These criteria include the delivery of approved documents and source code, peer review received, passing test results/up to task requirements as outlined, and sign-off by the project manager. Each milestone will be marked as complete only after acceptance criteria have been established, verified, and formally documented.

This systematic structure of schedule control addresses the need to manage the overall performance of the Ellmuna project to ensure deliverables are produced in time while maintaining accountability and transparency for the duration of the project life cycle.

3.3.4 Quality Control

The processes of quality control are essential in every project to maintain the advancement and have the best result over time, so we chose DMAIC approach to keep the same quality in each stage and control on it.

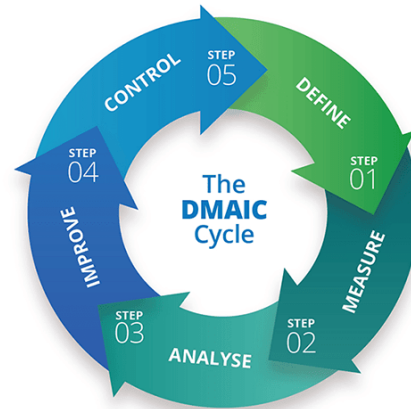


Figure 8 DMAIC Cycle.

DMAIC Cycle:

- Define: Determine and specify the issue and project goals.
- Measure: Gather data to figure out the recent rendition.
- Analyze: determine seed reasons for deficiencies or problems.
- Improve: Implement solutions to destroy the root issues.
- Control: Observe the advancement to sustain earnings.

3.3.5 Reporting

To facilitate open, comprehensive cycle-on-cycle communication about project direction, performance, and quality measures, the **Ellmuna** project establishes a structured reporting procedure. Reports will act as a formal mechanism of investigating requirements, schedule, budget, and deliverables to keep both internal and external stakeholders organized and informed.

Reporting Mechanisms & Information Flow:

The Project Manager generates and shares progress reports with all team members and external supervisors. The project reports summarize the status of completed tasks, schedule, budget, and risk assessments. Information will flow through two levels:

- **Internal Reporting:** Team members will share weekly and bi-weekly reports using Google Docs and Drive to facilitate collaborative tracking and transparency.
- **External Reporting:** The Project Manager will generate summarized project reports for submission to the course instructor or external supervisor. These reports will be submitted using the official submission mechanism in PDF format while following the SPMP reporting template.

Methods, Tools and Communication Techniques:

The team utilizes Microsoft Teams, Google Drive, and WhatsApp as the primary communication methods/tools. Google Drive holds all project documentation and report templates that will be easy to find and provides version control. Coordination in what we refer to as team “*huddles*,” and quick messaging updates occur in Teams and WhatsApp. Formal communication includes the use of email and **Google Docs** for written summaries of meetings, deliverable submissions, and quality reports. Routine meetings also include the recording of meeting minutes, capturing any decision-making or action items for follow-up.

Frequency and Level of Detail:

- **Weekly status reports:** Brief updates on progress, issues, and anything coming up in the following week.
- **Bi-weekly formal progress meetings:** Verbal and written summary of meeting includes performance metrics assessments and quality tracking.
- **Monthly summary reports:** Summary of everything tracked in the reports referring to each person's scope and project status for risk assessment and tracking progress/against schedule.
- **Final comprehensive report:** Submitted at project completion providing an overview of all accomplishments made, performance indicators during and completion of the project, and overall lessons learned and final thoughts.

All the reporting process will be within the scopes of the project, and with project visibility factor, ensuring the cycle of communication supports decision-making, accountability, and opportunities for continuous improvement throughout the **Ellmuna** project lifecycle.

3.3.6 Project Metrics

Project metrics are essential for quantitatively monitoring and controlling the progress, quality, and performance of the project throughout its lifecycle.

These metrics help the project manager and stakeholders evaluate whether the project is on schedule, within budget, meeting quality standards, and maintaining user satisfaction.

Data will be collected systematically using project management tools (e.g., Jira, GitHub, and Excel dashboards), and will be reviewed periodically to guide decision-making, identify deviations, and implement corrective actions promptly.

Metrics Collection Methods and Tools

Project metrics will be collected through both automated and manual tracking mechanisms.

- **Tools Used:** Google Sheets, GitHub Insights, Jira (for issue tracking), and internal testing reports.
- **Techniques:** Regular progress tracking, defect density analysis, velocity measurement, and quality assurance reviews.
- **Data Sources:** Project logs, code repositories, testing reports, and documentation feedback.

Table 0-11 Metrics Collection Methods and Tools

Metric	Description	Frequency of Measurement
Schedule Performance Metrics	Measures adherence to the project schedule, including task completion rate, schedule variance (SV), and milestone achievement to ensure timely delivery.	Weekly and at the end of each major phase.
Cost Performance Metrics	Monitors budget efficiency through budget variance (BV) and cost performance index (CPI), ensuring expenditures remain within the allocated financial plan.	Monthly and upon completion of each development stage.
Quality Metrics	Evaluates software quality through the number of detected defects, defect resolution rate, and user acceptance feedback to maintain system reliability.	After each major testing phase (unit, integration, user acceptance).
Productivity Metrics	Assesses team efficiency by tracking code output (e.g., lines of code per developer per week) and test case execution rate during development.	Weekly during development and testing phases.

Risk Metrics	Monitors the status of project risks by comparing the number of active risks against mitigated ones during each review cycle.	Biweekly during risk review meetings.
Customer Satisfaction Metrics	Measures user and supervisor satisfaction through feedback surveys and evaluation forms after major project milestones or final delivery.	After each key delivery (prototype, final version).

Validation, Analysis, and Reporting Processes

- **Validation:** Metrics data will be reviewed by the project manager and team leads to ensure accuracy and consistency before inclusion in reports.
- **Analysis:** Trends, variances, and correlations between performance indicators will be analyzed to detect potential risks or deviations.
- **Reporting:**
 - **Weekly reports** to the project team for progress updates.
 - **Phase-end reports** to the supervisor summarizing overall performance, including visual charts (progress, cost, and quality trends).
 - **Final report** documenting key project metrics and insights for future reference.

Purpose and Usage

These metrics serve as a foundation for continuous improvement, enabling the team to detect deviations early, optimize resource allocation, and enhance decision-making. The consistent collection and analysis of metrics will ensure project transparency, accountability, and measurable success.

3.4 Risk Management Plan

Potential risks can be deteriorating to the quality of a project. Thus, a detailed plan to deal with such risks in a timely and smooth manner is a must to minimize their effects. The main goal of risk management is to identify the risk and manage it by creating plans to avoid, mitigate, or deal with it. Risks may frequently arise during the project, so the management should create a clear plan that addresses most possible scenarios. They should also estimate the likelihood of each risk and comprehend how it will affect the project. In addition, hazards don't have the same influence throughout project implementation. Some of them might vanish, while others might grow.

The manager must therefore constantly keep an eye on these hazards to act immediately. The following plan outlines our approach to managing risks for the **Ellmuna** app project.

3.4.1 Risk Identification and Classification

Risks are identified during project initiation and revisited periodically throughout development. The project team conducts brainstorming sessions, stakeholder interviews, and project document reviews to identify both internal and external risks. Risks are classified into categories such as:

- **Technical Risks:** issues with new technologies, software compatibility, or integration failures.
- **Project Management Risks:** poor estimation, scope creep, or schedule slippage.
- **Operational Risks:** inadequate testing resources, data loss, or communication failures.
- **External Risks:** changes in university policies, internet connectivity issues, or stakeholder availability.

Each identified risk is documented in the Risk Register with a unique ID, description, probability, impact, owner, and mitigation plan.

3.4.2 Risk Tracking and Reporting

Risk tracking will be maintained using a Risk Register Spreadsheet, updated weekly by the project manager. The status of each risk will be reviewed in team meetings. Key updates include new risks, closed risks, or changes in probability/impact levels.

Reporting Procedures:

- **Weekly Reports:** Summarize current high-priority risks and mitigation actions.
- **Milestone Reviews:** Evaluate new risks introduced in each phase transition.
- **Final Report:** Include summary of risk outcomes and lessons learned.

3.4.5 Risk Management Roles and Responsibilities

Table 0-12 Risk Management Roles and Responsibilities

Role	Responsibilities
Project Manager	Oversees all risk management activities, maintains the risk register, and ensures timely reporting.
Development Team Lead	Identifies technical risks, monitors progress, and applies mitigation strategies.
Quality Assurance Lead	Detects testing-related risks, evaluates impact, and reports test delays or failures.
Client/Supervisor	Reviews major risks and approves mitigation and contingency plans when applicable.

3.4.6 Communication of Risks

All risk communications will be transparent and traceable through:

- **Email updates** for critical risk alerts.
- **Shared Google Sheet** for live risk register tracking.
- **Weekly progress meetings** for reviewing mitigation actions.
- **Supervisor briefings** for escalation of high-impact risks.

3.4.6 Schedule for Risk Management Activities

Table 0-13 Risk Management Activities

Activity	Frequency	Responsible Party
Initial Risk Identification	At project initiation	Project Manager & Team
Risk Assessment	Bi-weekly	Project Manager
Risk Register Update	Weekly	All Team Members
Risk Review Meeting	Every two weeks	Entire Team
Final Risk Review	At project closure	Project Manager

3.5 Project Closeout Plan

The **Ellmuna** project involves a systematic and organized closeout approach to complete the project in a documented manner. The closeout plan specifies how team members will be reassigned, future documentation will be archived, individuals will be provided for each member's performance, and a final summary of the project will be prepared.

Staff Reassignments: Following the final approval of intended project deliverables, the project team will be reassigned to new academic or project duties in the Computer Science Department. The project manager will be responsible for ensuring that adequate time is allocated for all project-related activities to cease prior to reassigning any member. Each member will summarize all remaining project responsibilities for the project team and submit their files for all relevant project activities prior to the closeout of the project.

Archiving of Project Files: All project artifacts—source code, design documentation, meeting notes, reports, and presentations—will be retained in a shared Google Drive repository maintained by the team to ensure long-term and future accessibility for continuation of academic and departmental projects. A backup copy of all relevant files will remain in the instructor's submission repository for recordkeeping purposes.

Collecting Project Data: Once the project has been completed, the numerical data related to performance (schedule adherence, task completion rate, frequency of issues, quality of outcomes, etc.) will be summarized and entered into the department's business projects database, allowing for evaluation of the effectiveness of the **Ellmuna** project's approach to project management and serving as baseline data for future course projects.

Post-Mortem Debriefings: A final debriefing will be conducted with the membership to reflect upon the overall process. The debriefing will allow the membership to address aspects of the project that went well, identify issues that arose, and share opportunities for improvement for future projects. Notes from the post-mortem debriefing will be recorded and provided to the instructor as part of the final project deliverables.

Final Report and Lessons Learned: The Project Manager will compose a final report for the project, and all members will be provided an opportunity to review it. The report will discuss what was accomplished in the project, discuss whether appropriate objectives were met, and summarize lessons learned during the project. The report will also include performance data analysis and recommendations for enhanced project success in any future semester.

Finishing these closeout activities will ensure that the **Ellmuna** project can be closed in a professional and orderly manner, retaining all outcomes, lessons learned, and documentation for future references and the improvement process.

Risk Management Process

The Risk Management Process involves five key steps that help the project team identify, assess, and control potential risks throughout the project lifecycle:

1. Identify:

In this stage, the team recognizes possible risks that could affect the project's objectives. Risks can be technical, organizational, or external factors that may impact progress or outcomes.

2. Analyze:

Once identified, each risk is analyzed to determine its likelihood and impact. This helps the team understand which risks are more critical and require immediate attention.

3. Prioritize:

Based on the analysis, risks are ranked from highest to lowest importance. The team focuses resources on addressing the most severe risks first.

4. Mitigate:

Appropriate strategies are developed and implemented to reduce or eliminate the impact of high-priority risks. This may include contingency plans or preventive actions.

5. Monitor:

Throughout the project, the team continuously monitors existing risks and identifies new ones. Regular updates ensure that mitigation measures remain effective and relevant.

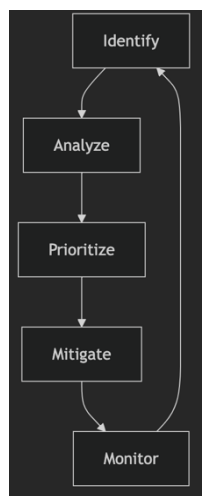


Figure 9 REM cycle

Key Risk Management Components

Table 0-14 Main parts of our risk management plan and what they do

Component	Description
Contingency planning	Develop strategies for top 5 risks: regular testing and updates
Risk Tracking	Centralized register; assigned risk owners; weekly status updates
Risk Level changes	Predefined thresholds; rapid response for critical risk
Work Activities	Identification, assessment, prioritization, mitigation, monitoring
Documentation	Digital register; weekly, monthly, and quarterly reports
Communication	Regular updates; clear channels with all stakeholders

Risk Management Schedule

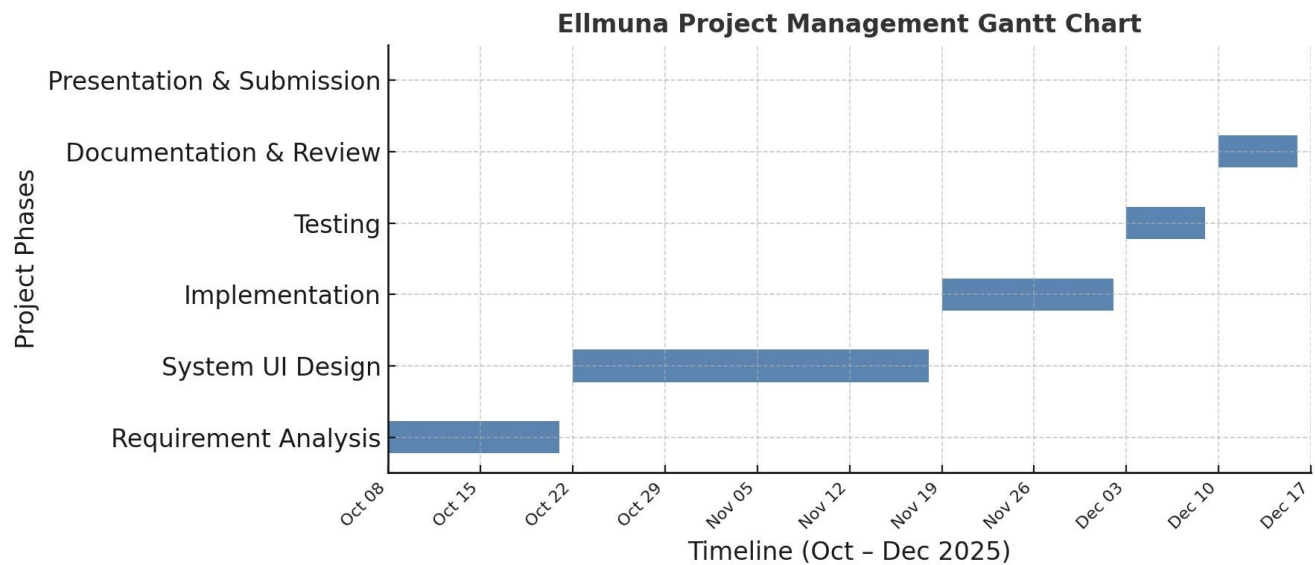


Figure 7 Management Timeline

Responsible Personnel

- **Project Manager:** Overall risk management oversight
- **Risk Coordinator:** Day-to-day risk management activities
- **Team Leads:** Risk identification and mitigation within their areas
- **All Team Members:** Continuous risk monitoring and reporting

Identified Risk Factors and Their Potential Impact

Table 0-15 Risk Category

Risk Category	Examples
Customer-Project Relationship	Miscommunication of Zakat calculation requirements; Changes in client's organizational structure
Contractual	Unclear deliverable specifications; Disputes over intellectual property rights
Technological	Integration challenges with existing financial systems; Security vulnerabilities
Product Complexity	Underestimation of calculation complexities; Scalability issues
Development Environment	Compatibility issues between tools; Inadequate testing environment
Personnel	Difficulty finding skilled developers; Potential loss of key team members
Schedule and Budget	Delays due to unforeseen complexities; Budget overruns
Customer Acceptance	Misalignment between features and expectations; Resistance to adoption

By implementing this comprehensive risk management plan, we aim to proactively identify and mitigate potential issues, ensuring the successful delivery of our **Ellmuna** app project.

3.5 Project Closeout Plan

As the **Elmuna** app project approaches completion, it is essential to have a well-structured closeout plan. This plan ensures a smooth transition, captures valuable insights, and properly concludes all project activities.

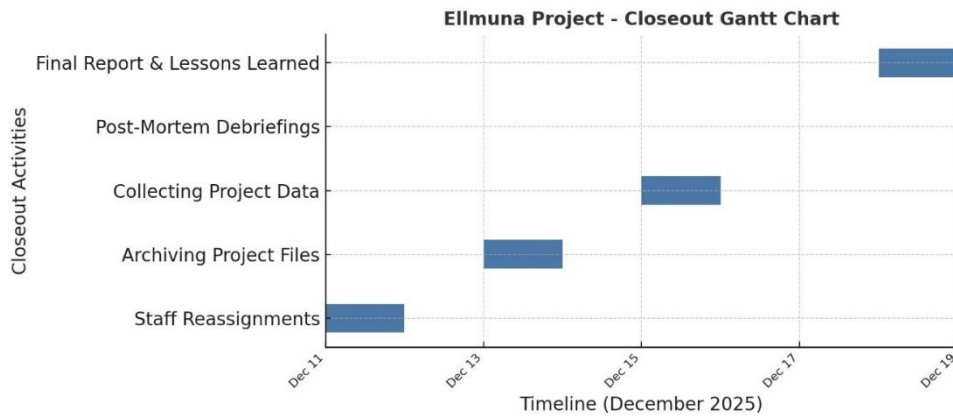


Figure 8 Closeout Timeline

This plan ensured the effective conclusion of the **Elmuna** app Project, preserving knowledge and setting the stage for future Islamic finance software innovations

4. Technical Process Plans

4.1 Process Model

The Agile software development methodology was adopted for the **Ellmuna** project because it provides flexibility, continuous improvement, and close alignment with user needs. The process consists of multiple short development iterations (Sprints), each focusing on a specific system feature such as GPA calculation, university directory, or user interface design.

Relationships Among Major Activities

Each sprint includes the following iterative phases, which are connected sequentially:

1. **Planning and Analysis:** Identify student needs, gather requirements, and define the functionality of the sprint. Input: User feedback, academic data requirements. Output: Updated requirement document.
2. **Design:** Create UI wireframes and database schemas using Figma and Firebase, ensuring consistency with previous iterations. Output: Design prototypes, database diagrams.
3. **Implementation:** Develop the sprint's functionality using Java and Flutter within Android Studio. Each module is integrated with existing components. Output: Functional module ready for testing.
4. **Testing:** Verify system accuracy—especially GPA and weighted score calculations—through manual and automated testing. Output: Test reports, bug logs, and fixes.
5. **Review and Improvement:** Present sprint results to the team and supervisor, gather feedback, and plan improvements for the next sprint. Output: Sprint review report and improvement plan.

Flow of Information and Work Products

Information flows continuously between team members using Google Drive and WhatsApp group discussions. Each sprint's outputs become inputs for the next sprint (e.g., tested modules from Sprint 1 feed into Sprint 2 for integration).

Timing of Work Products

Each sprint lasts approximately 2 weeks, producing specific deliverables such as design prototypes, code modules, and updated documentation. These deliverables are reviewed at the end of each sprint.

Reviews and Approvals

- Sprint (iteration) Reviews: Conducted at the end of each sprint to evaluate progress.
- Supervisor Review: Conducted after major milestones (Design, Prototype, Final Product).
- Team Peer Review: Members cross-check code and documents for consistency.

Major Milestones

1. SRS Approval – End of Requirement Analysis
2. Prototype Completion – After 3rd Sprint
3. Testing Completion – Before Final Revision
4. Final Presentation – December 2025

Baselines

- Requirement Baseline: Established after the SRS approval.
- Design Baseline: Set after the design of UI and database structures.
- Code Baseline: Maintained at the end of each sprint via GitHub.

Project Deliverables

- Fully functional Android application **Ellmuna**
- Documentation: Proposal, SPMP, SRS, SDD, STS, Final Report, User Manual
- Presentation slides and demo video.

Required Approvals

All sprint outcomes and documentation must be approved by Dr .Noorah Al-Behaijan before moving to the next development stage.

Process Model Diagram

The figure below illustrates the cyclical Agile workflow for the **Ellmuna** project ,showing how the process flows from Planning to Design ,Implementation ,Testing ,and Review before looping back to Planning

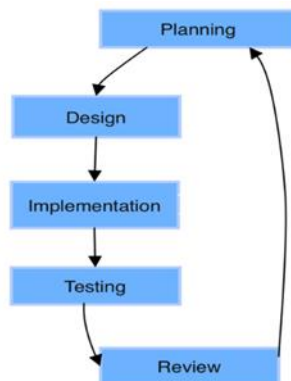


Figure 12 Agile Process Model for Ellmuna Project

Tailoring the Standard Process

The standard Agile model was slightly tailored to fit academic project constraints—shorter sprints , combined documentation phases ,and direct supervisor feedback after each sprint instead of a formal product owner review.

4.2 Methods, Tools, and Techniques

The **Ellmuna** project follows the Agile Iterative Development methodology, which allows for flexibility, continuous improvement, and user-centered feedback at the end of each iteration. Each iteration covers the phases of specification, design, implementation, testing, integration, and documentation.

Programming Languages and Frameworks

- Java: Used for developing backend logic and core calculation functionalities.
- Dart with Flutter Framework: Used for building the mobile application interface and ensuring cross-platform compatibility.
- SQL (Firebase / Cloud Firestore queries): Used to manage and query stored data related to users, GPA records, and university information.
- HTML, CSS (for web demo): Used to design and test UI prototypes before mobile integration.

Processes, Tools, and Techniques

Table 0-21 Tools and Techniques Used in Each Project Phase

Phase	Tools / Techniques Used	Purpose
Requirement Specification	Google Docs, MS Word	Writing and revising the SRS document
System Design	Figma, Draw.io (UML Diagrams)	Designing UI and architecture
Implementation	GitHub	Coding and version control
Database Integration	Firebase , SQL	Managing authentication and data storage
Testing	Manual and Functional Testing	Verifying accuracy of GPA and score calculations
Documentation	MS Word, Google Drive	Preparing reports and storing shared files
Collaboration	WhatsApp	Team communication and coordination

Technical Standards and Policies

- IEEE Standards.
- Coding Standards: Meaningful naming conventions, modular structure, and proper commenting in code.
- Documentation Standards: Version-controlled documents stored in Google Drive.
- Review Procedures: Supervisor reviews after each iteration to ensure compliance with academic and technical standards
- Data Policies: Use of Firebase with secure read/write permissions to protect student data.

Maintenance and Modifications

Post-development modifications are managed through version control (GitHub). (Any future maintenance or updates will follow the same Agile iteration process, ensuring continuous enhancement without disrupting system stability.

4.3 Infrastructure

The **Ellmuna** project was developed using simple and available tools that support all stages of building, testing, and improving the system. The development environment includes the following components:

Hardware

- Personal laptops used by team members, each with at least 8 GB RAM.
- Android phones are used to test the mobile application on real devices.
- personal computers used for preparing documents and presentations when needed.

Software

- Android Studio: main tool used for programming and testing the application.
- Visual Studio Code: used for editing project files and scripts.
- Firebase Console: used to store user data and connect the app to the cloud.
- GitHub: used to save project versions and share code among team members.
- Google Drive: used to share and store project reports and documents.
- Figma: used to design the app interface before starting the coding process.

Network

- The project requires a stable internet connection to connect with Firebase and synchronize project files on GitHub.
- All development and testing are done locally, and then the data is synced with Firebase in the cloud.

Policies and Security

- Each team member works on their own device with secure access.
- GitHub is used to control and manage changes to the code.
- Firebase provides data security and backup for all stored information.

Facilities and Support

- Work was mainly done from home and sometimes in the Department labs.
- Team members communicated through Google Meet and WhatsApp for meetings and updates.

The project supervisor, Dr. Noorah Al-Behaijan, provided continuous guidance and feedback.

Maintenance

The team will keep maintaining **Ellmuna** using GitHub and Firebase by fixing bugs, adding new features and keeping all data updated and safe.

The acceptance of the **Ellmuna** system will be based on confirming that the final product meets both the technical and academic requirements defined during the project. The acceptance process will be carried out by the project supervisor and student evaluators through testing, review, and demonstration sessions.

Formal testing and evaluation sessions will be conducted at the end of the final iteration .During these sessions ,the supervisor and selected student users will test the system's main functions ,evaluate performance ,and ensure that all planned features are working as expected .Any issues found will be documented and corrected before final approval.

Acceptance Criteria

The product will be accepted if the following conditions are met:

- GPA and weighted score calculations are accurate for both the 4.0 and 5.0 grading systems.
- The interface is user-friendly and easy to navigate, confirmed through usability testing with high school and university students.

-
- The application runs smoothly and efficiently on Android devices with no major crashes or bugs.
 - All planned features (GPA calculator, weighted score tool, and university directory) are fully functional.
 - The system receives positive feedback from evaluators and the academic supervisor.

Formal Agreement

The acceptance of the project will be confirmed through written approval and evaluation by the supervisor, Dr. Noorah Al-Behaijan, representing the Computer Science Department Jubail campus at Imam Abdulrahman bin Faisal University. This serves as the official confirmation that the project meets the academic and technical expectations.

Testing and Validation Methods

- Functional Testing: Verifying the accuracy of GPA and weighted score calculations
- User Testing: Conducting trials with students to assess ease of use.
- Performance Testing: Ensuring stable and fast response times.
- Supervisor Review and Demonstration: The final demonstration will include a live walkthrough of all features and project documentation for final validation.

5. Supporting Process Plans

5.1 Documentation

This section discusses the documentation plan that was applied to the **Ellmuna** project. All project documents were collaboratively created by the entire team using course templates and incorporating IEEE standards where applicable.

Table 0-22 Documentation

Phase	Submission Date	Approximately number of pages	Format Standard	Document preparation	Document Review
Project Proposal	27 Sep.	5	Supervisor	All members	Dr. Noorah Al-Behaijan
Project Management Plan (SPMP)	18 Oct.	60	IEEE Std. 1058-1998		
Project Requirements (SRS)	27 Oct.	15	IEEE Std. 1058-1998		
Status Report	17 Nov.	5	IEEE Std. 1058-1998		
Project Design (SDS)	03 Nov.	50	IEEE Std. 1058-1998		
Project Test Plan (STS)	24 Nov.	40	IEEE Std. 1058-1998		
Projects Presentations	15 Dec.	20	Supervisor		

Documentation Control and Standards

- All documents follow templates provided in the course with IEEE standards.
- Title page, revision history, version number, and authors list, are included in the document.
- Files are shared in a Google Drive folder for version control and teamwork collaboration.
- The final versions are submitted through the Blackboard system as PDF files.

Review and Approval Process

- Documents are collaboratively drafted by the team
- The team collectively reviews and edits for accuracy and consistency.
- The team leader performs a final review
- The approved version is submitted to the supervisor for evaluation.

6. Additional Plans

This section provides additional plans that ensure consistency of the application quality and ease of use after submitting it. The project adheres to the standards of security and privacy by not collecting or storing any personal data of users, and the academic data entered manually by the user is sufficient. All the data will be saved inside the user's device without the need to send to any external servers, to ensure user privacy and complete protection of the user's data. The application interface is designed in a simple, interactive, and clear way that does not require any prior technical training. In addition to providing a user manual that explains detailed steps, accompanied by screenshots to help users enter data and calculate the weighted average or GPA, and easily browse available universities and majors. It will also follow a maintenance plan after submitting the final version of the application. Additionally, it includes reviewing and updating the universities and majors' data constantly. Fixing bugs and improving performance based on user feedback. Furthermore, update the programming libraries used when newer versions are released. As for the support plan, the users will be able to communicate with the development team via email or through the project page on GitHub to report any problems or provide feedback. The responses will be within a period that does not exceed 48 working hours. The plan also includes an integration plan that aims to link the application in the future with official academic resources. Or Saudi universities' websites via the application programming interfaces (APIs) to update information about the majors, and the acceptance rates are calculated automatically without the need for manual entry.